Eman Hosam 16

Exercises

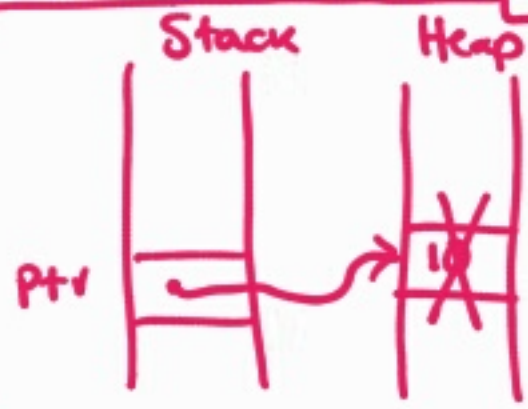
1

```
int * ptr = new int;
*ptr = 10;
delete ptr;
```

«Unexpected Behavior»
↳ Runtime Error: Access violation

```
*Ptr = 20; // Cout << *pto;
```

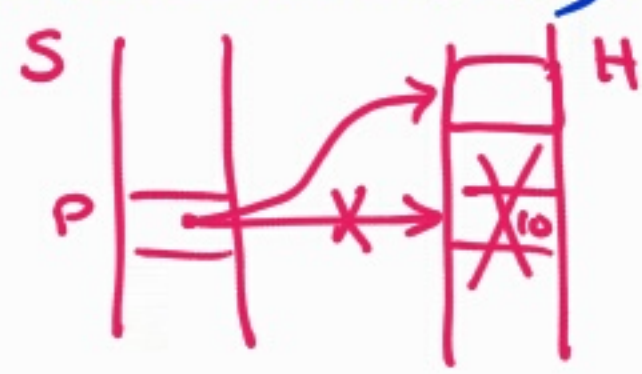
«Unexp. B»
↳ 10
↳ garbage



2

```
int * p = new int;
*p = 10;
delete p;
p=new int[];
```

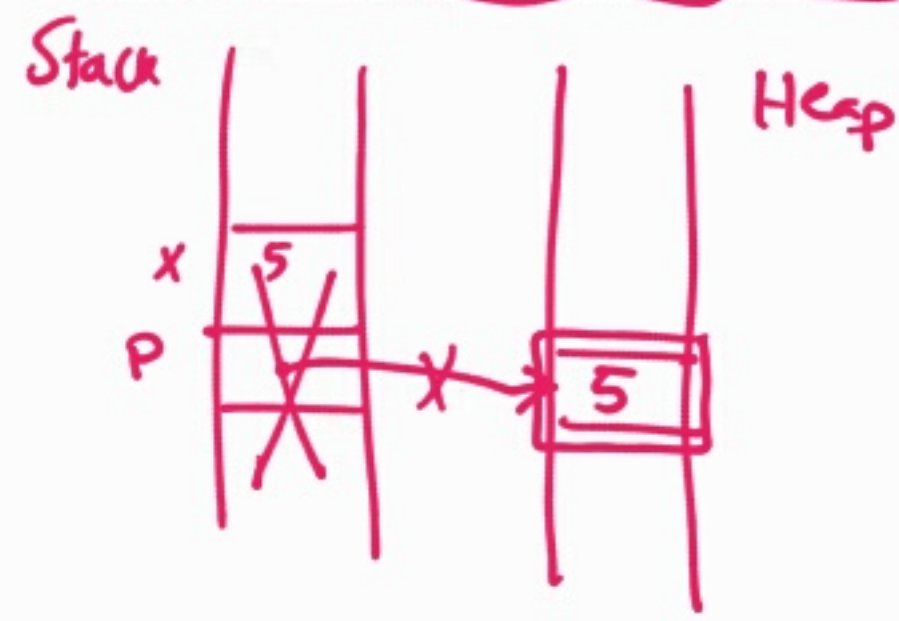
«Unexpected Behavior»



3

```
int* q;
if ( ... )
{
    int * p = new int;
    *p = 5;
    // delete p; ↔ // q = p;
}
// dealloc of p
int x = 5;
⋮
delete q;
```

at least 1 pointer بيشاور على ال new int عشان توصلها



Waste of memory

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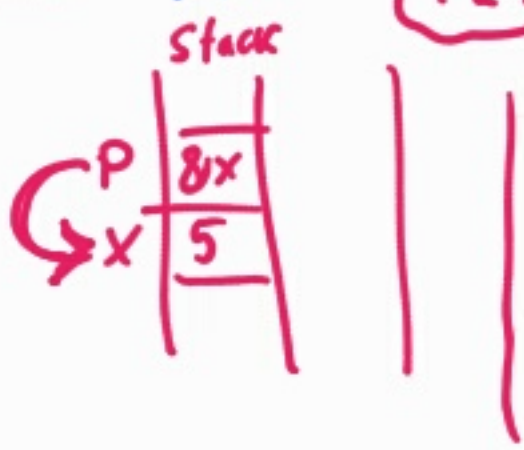
4

```
int x = 5;
int *p = &x;
delete p;
```

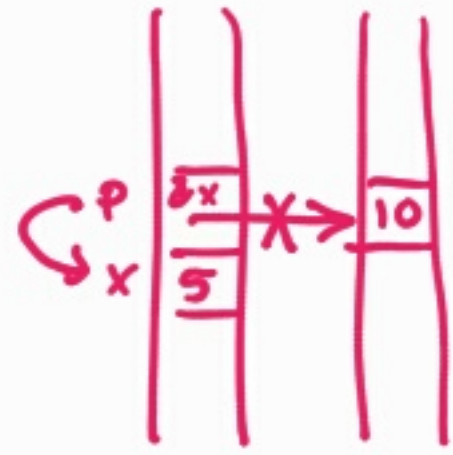
// Runtime Error
العجيب



5

```
int x = 5;
int *p = new int;
*p = 10;
p = &x;
delete p;
```

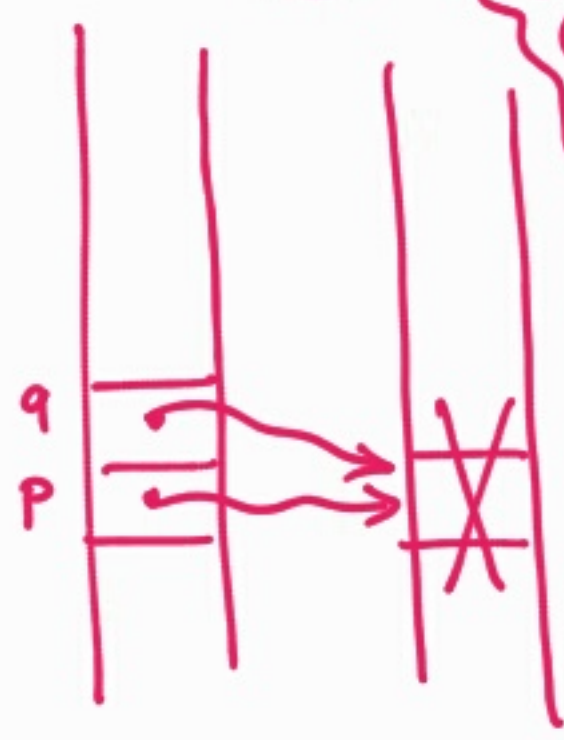
// Runtine Error
العجيب
نفس ال



6

```
int *p = new int;
if (...)
{
  int* q = p;
  delete q;
}
delete p;
```

double delete
نفس ال
Runtine Error
العجيب



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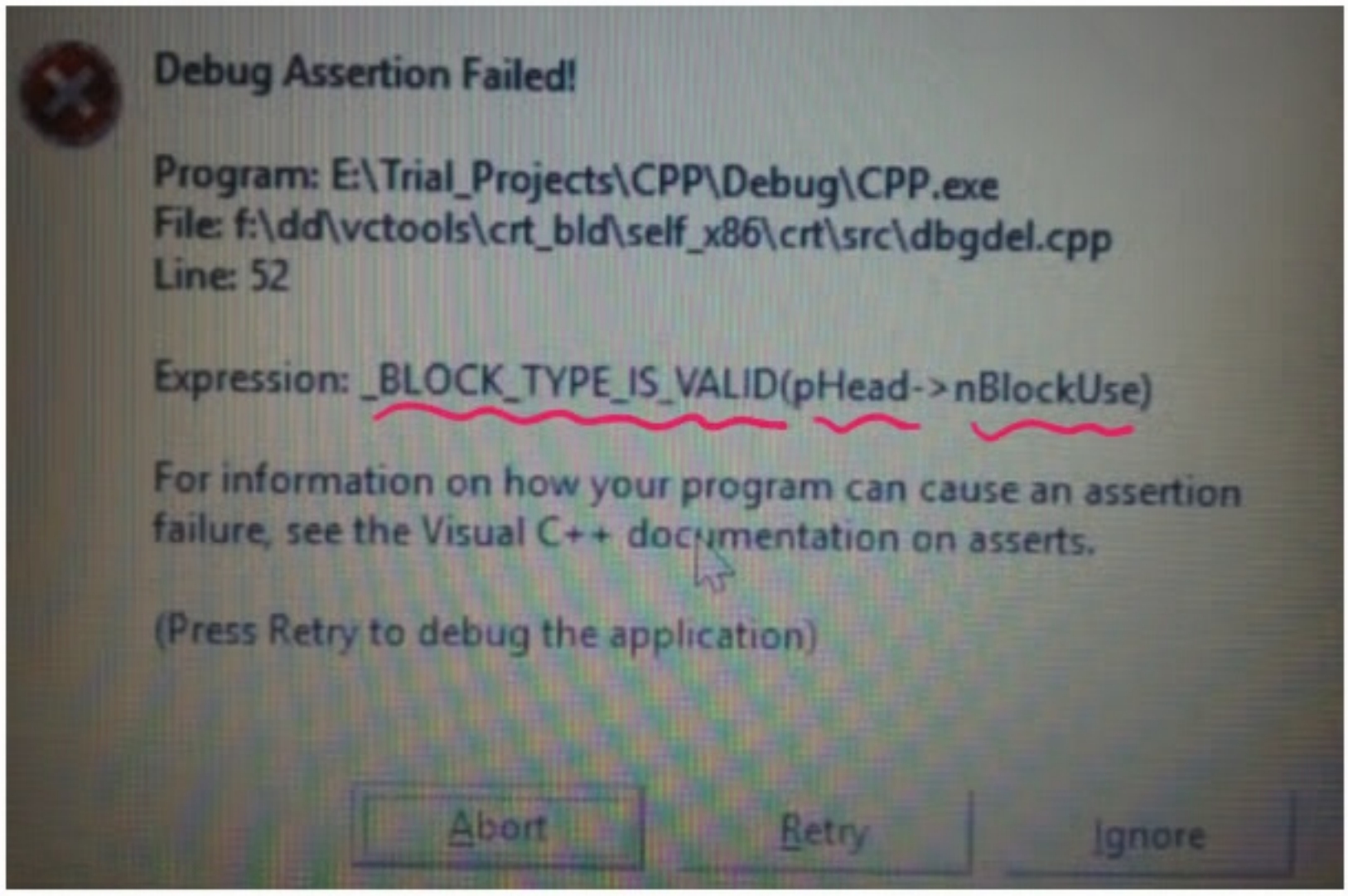

① delete an automatic var.

② delete memory bc. already deleted.

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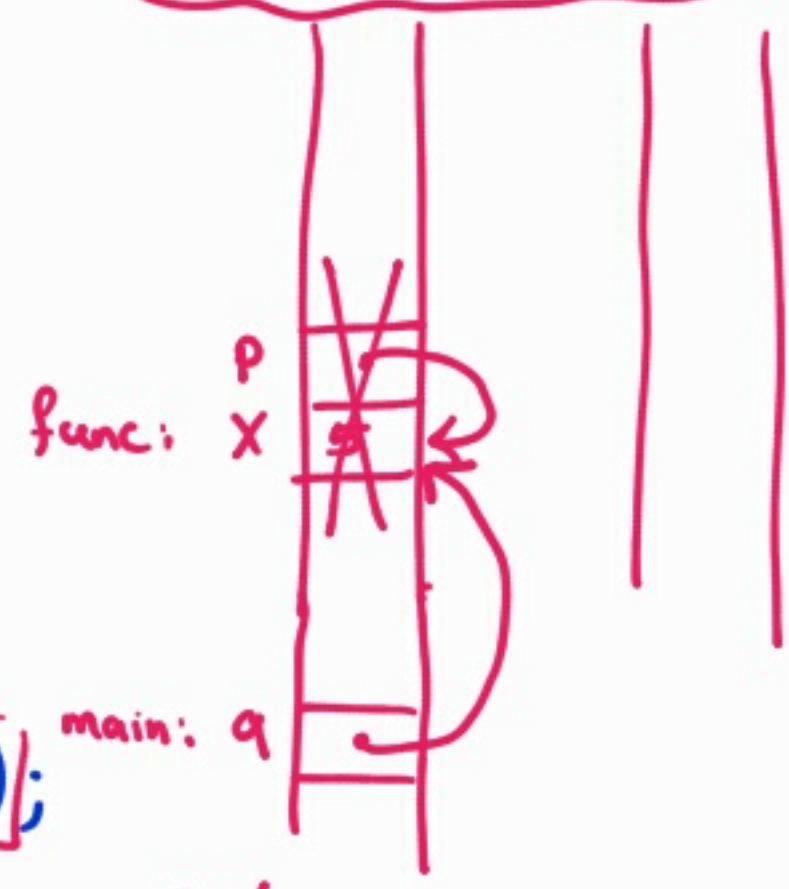
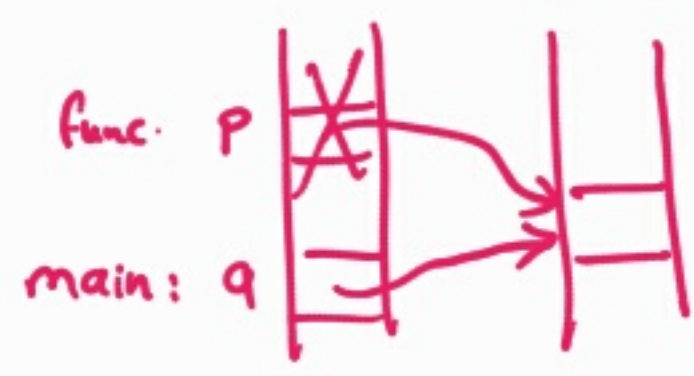
```
int* func()
{
    int x = 5;
    int* p = &x;
    return p;
}
int main()
{
    int* q = func();
    *q = 5;
}
```

Never return address of an auto. var



⟹ unexpected behavior ⟹ Access Violation Runtime Error

```
int* func()
{
    ~~int x = 5;~~        int* p = new int;
    ~~int* p = &x;~~
    return p;
}
int main()
{
    int* q = func();
    *q = 5;
    delete q;
}
```

→ return of address of dyn. alloc. mem.